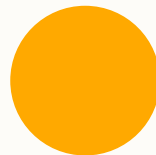




Networking Layers

Reese Hatfield



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Networking

- We've discussed a network conceptually
- But how do they *actually* communicate?
- Wires?
- Wireless?
- Standards?
- Protocols?





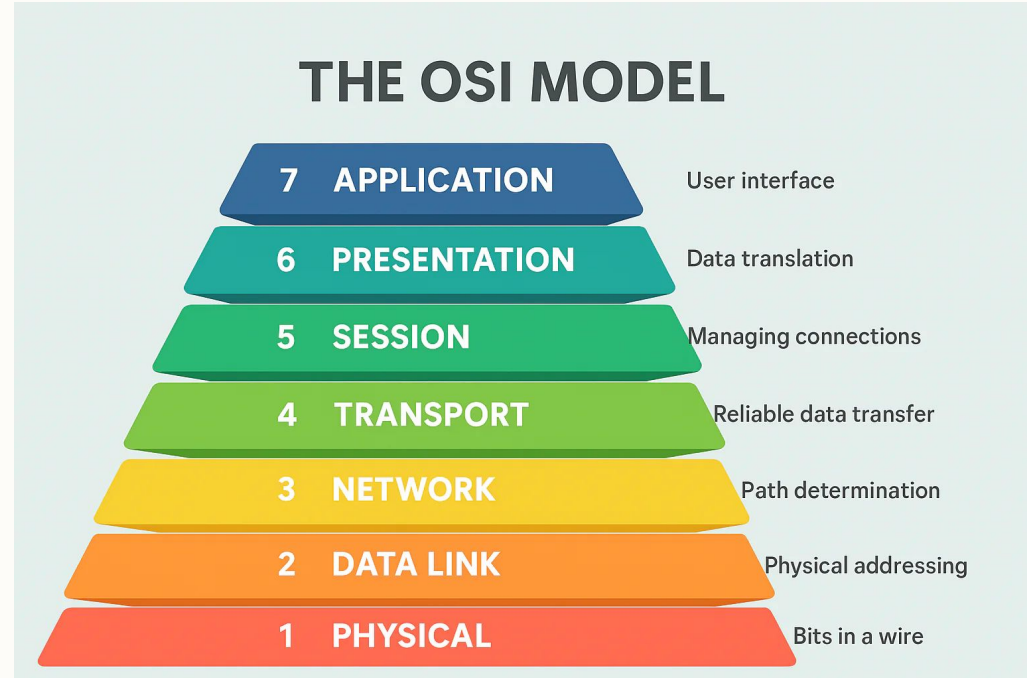
Networking

- **OSI** standard model for networking
- Breaks networking down into layers of abstraction
- From the user interface (e.g. web browser)
- To your wifi standard
- To bits in a physical wire



Networking

- **7 Layers**
- Under the hood of one protocol, lies another
- Until we get bits





Networking

- Different implementations of all of these different layer
- One implementation is not bound to any other
- Networking generally occurs when all of these have provided some implementation*





Networking

- Layer 7
- **Application Layer**
- **Top Layer**
- Direct interface between the user facing app and the network
- HTTP exists at layer 7





Networking

- Layer 6
- **Presentation Layer**
- Converts data and ensures each end of the network can read the data
- ***JSON***, ASCII, XML, etc.





Networking

- Layer 5
- **Session Layer**
- Manages network sessions
- Opens, closes, and restarts sessions
- Security Handshakes





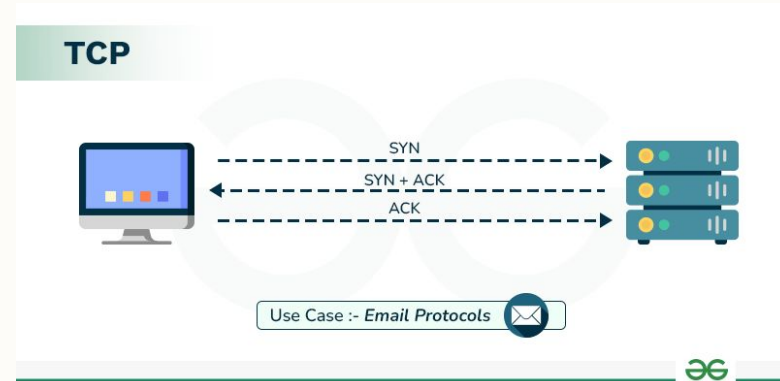
Networking

- Layer 4
- **Transport Layer**
- **Important one, spend some time here**
- Manages how data is delivered within a network
- Two common protocols **TCP** and **UDP**



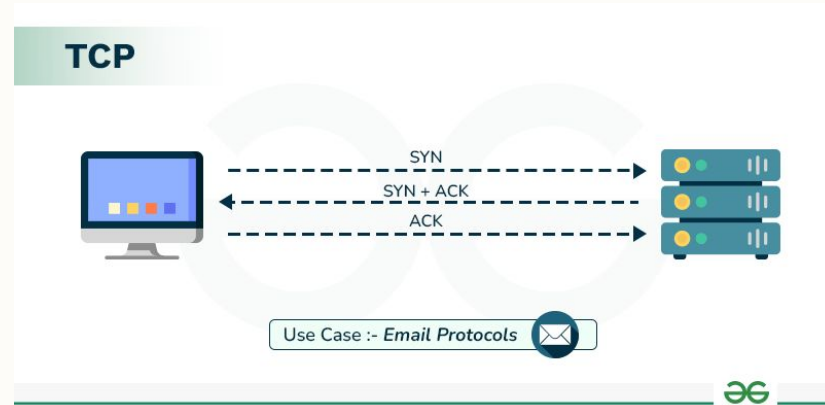
Networking

- TCP
- Transmission control protocol
- Three way handshake
- SYN
- SYN-ACK
- ACK



Networking

- Handshake
- A bit directional way to ensure there is a pre-existing stable Connection
- Synchronize
- Acknowledge



Networking

- **UDP**
- User Datagram Protocol
- No handshake involved
- Faster, less reliable
- Can be swapped





Networking

- Layer 3
- **Network Layer**
- IP Addressing, send ***packets*** around
- Send this packet to this IP address
- But what's in a packet?





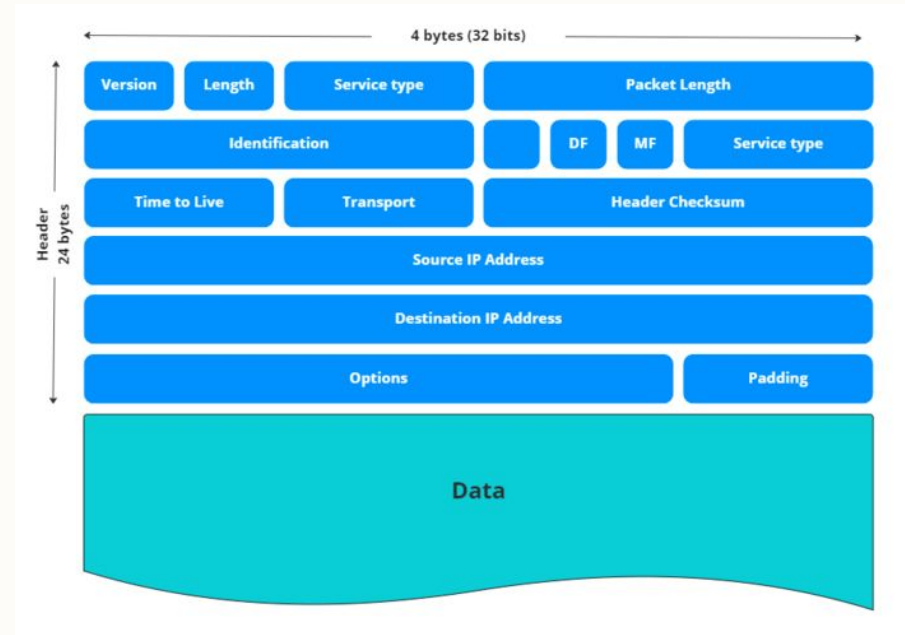
Networking

- **Packet**
- Logical unit of networking
- Contain information about the packet
 - Source
 - Destination
 - Type, etc
- As well as arbitrary data



Networking

- **Packet Structure**
- Corresponds to a series of bits
- Standardized
 - Handshakes
 - Information, etc





Networking

- Layer 2
- **Data Link Layer**
- Device to device data transfer
- Direct link between devices
- Responsible for **physical addressing**
- How are devices addressed?





Networking

- **MAC Address**
- Media Access Control Address
- 12 Digit Hex ID
- Assigned by the manufacturer
- Unique and Permanent for every device
 - **1a:2b:3c:4d:5e:6f**





Networking

- Layer 1
- **Physical Layer**
- **Bottom Layer**
- Raw transmission of bits over physical wires (usually*)
- Works at the level with gates

